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Promoting College Student's Learning Motivation and Creativity through a STEM Interdisciplinary PBL Human-Computer Interaction System Design and Development Course

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ABSTRACT

In recent years, STEM (Science, Technology, Engineering, and Mathematics) has been extensively advocated and implemented in education, as it is suggested to be very impactful on student's interdisciplinary learning, which can be seen as a significant driving force for a country's advancement in scientific and technical knowledge, innovation, economy, and international competitiveness. Developing a human-computer interaction (HCI) system to solve real-world problems requires the inventors to have interdisciplinary STEM knowledge and skills, thus a STEM Interdisciplinary Project-based Learning (IPBL) approach was applied to teach a total number of 45 college students registered in the departments of engineering and design. Inspired by Design Thinking, the 18-week STEM IPBL course was delivered through four phases,

including discover, define, develop, and deliver. All the accomplished HCI projects applied the interdisciplinary knowledge and skills from the domains of STEM. Evidence drawn from the 6-point Likert 'Motivated Strategies for Learning Questionnaire (MSLQ)' indicated that the STEM IPBL course was very impactful on student's learning, which improved the participants' (a) overall learning motivation (Pre M= 4.4, Post M= 4.64; $p = .012$), (b) self-efficacy of learning (Pre M= 4.03, Post M= 4.43; $p = .003$), (c) enjoyableness of learning STEM (Pre M= 4.68, Post M= 4.75; $p = .556$), and (d) recognizing the significance of learning STEM on future career development (Pre M= 4.73, Post M= 4.94; $p = .077$). It is also found that compared with design majored students, the course had a better effect on the engineering majored students. Evidence collected from 'Abbreviated Torrance Test for Adults (ATTA)' indicated that the student's overall creativity was significantly improved (Pre M= 63.36, Post M= 68.44; $p = .000$). More specifically, among the four facets of creativity, the improvements were as follows: fluency (Pre M= 14.89, Post M= 16.2; $p = .001$), elaboration (Pre M= 16.69, Post M= 18.62; $p = .000$), flexibility (Pre M= 14.82, Post M= 16.04; $p = .009$), and originality (Pre M= 16.96, Post M= 17.58; $p = .136$). It is found that the STEM IPBL course had a different impact on the student's originality, while the originality of engineering majored students significantly improved ($p = .006$), the originality of design majored students did not change. Some educational implications were also provided in the article.

Keywords: STEM education; Project-based Learning (PBL); Human-Computer Interaction (HCI) system; Creativity; Engineering Education

1. INTRODUCTION

The development of Human-Computer Interaction (HCI) system has received increased attention. HCI system development often requires interdisciplinary knowledge, such as design and technology (Hartson, 1998; Hewett et al, 1992; Preece et al, 2015; Te'eni et al, 2007), persuasive technology for making behavior change (Agapie et al, 2016; Consolvo et al, 2009; Fogg, 2002; He et al, 2010; Hekler et al, 2013) or requires interdisciplinary team collaborations (Duncan et al, 2014; Lathia et al, 2013; Stawarz, 2015) to optimize the requirements of desirability, feasibility and viability (Kandachar, 2010; Smith, 2007) for developing products, services and technologies for HCI systems.

In recent years, STEM has been extensively advocated and implemented in education to nurture learner's interdisciplinary knowledge and skills, which can be seen as a significant driving force for a country's development in scientific and technical knowledge, innovation, economy, and international competitiveness. STEM education has thus emerged as a paramount goal of many countries in the pursuit of excellence. STEM refers to "Science", "Technology", "Engineering", and "Mathematics", which is the skill set that is "expected to be held by people with a tertiary-education level degree in the subjects of science, technology, engineering and mathematics" (EU Skills Panorama Glossary, 2014; Fans and Ritz, 2014). Project-based learning (PBL) has also been widely recognized as an effective approach to immersing the learners in a contextualized and authentic learning setting, has therefore been commonly utilized to nurture the learner's creative problem solving capability in dealing with real-world problems (Grant and Branch, 2005; Hsu, 2004; Scott et al, 2004). Nonetheless, it should be noted that it is quite challenging for engineering students to conduct an engineering project and make a product that is well-functioned and attentively addresses the demands of target users under PBL teaching method. Thus, for many engineering

education, courses that require great amounts of hands-on experience extensively adopt PBL to cement students' practical use of knowledge (Calvo, Cabanes, Quesada, & Barambones, 2018).

The key issue that alarms us is that developing a useful and satisfactory HCI system development requires interdisciplinary knowledge and a set of skills from the areas of engineering, computer science, and design. While the expertise of engineering and computer science can help develop a system with adequate functions, the expertise of design can help improve the understanding of target users, better product ergonomics, and aesthetics, and even redesign or create the services associated with the products. Therefore, a STEM-based interdisciplinary PBL (STEM IPBL) has been proposed in this study, in which students from engineering and design departments were strategically grouped to work collaboratively on HCI system development projects, and helped them to acquire interdisciplinary STEM knowledge and skills more effectively. Besides the commonly used PBL approach in other studies, the STEM IPBL approach used by the current study has several features, including (a) The course taught throughout the experimental teaching placed a heavy weight on the learning and applying of STEM interdisciplinary knowledge and skills to solve more complex problems (Dym et al, 2013). The details of the course design that systematically integrated the knowledge and skills of STEM can be referred to Table 1. (b) Students from different majors were asked to be grouped for HCI systems development (each group contains 1-2 design majored student(s) and 2-3 engineering majored students. While the expertise of engineering and computer science can help develop a system with adequate functions, the expertise of design can help improve the understanding of target users, increase product ergonomics and aesthetics.

The training of both design thinking and the ability to apply engineering knowledge into practice is systematically embedded in the STEM IPBL course (Cropley, 2016; Brown, 2009). All the groups are required to use their creative problem-solving skills to accomplish HCI system project that can be used to solve real-world issues. Furthermore, the basis of working in designated task groups enable the group members to work cooperatively and collaboratively whilst applying interdisciplinary STEM knowledge and skills in HCI system development for alleviating real-world problems. Thus, it is reasonable to expect that through the collaboration process within the cooperative learning, the students' communication and social skills can also be improved by participating in the course (Hirsch et al, 2001; Mamykina et al, 2002).

2. METHOD and MATERIALS

2.1 Participants

A total number of 45 participants aged 22-24 years old were recruited from a top tier university in Taiwan. All the participants were registered as undergraduate and graduate students studying in the (1) 30 students registered in engineering departments (Computer Science and Information Engineering: CSIE, and Electronic Engineering: EE) and (2) 15 students registered in design department (Industrial Design: ID). In total, 14 groups (3-5 participants per group) were formed to undertake their specific HCI system projects. It should be noted that in order to stimulate the interdisciplinary collaboration and communication in the designing and development of HCI systems, each group was asked to be formed by students of different majors, namely 1-2 design majored student(s) and 2-3 engineering majored students. It was expected that the expertise of engineering and computer science can help develop a system with adequate functions, whereas the expertise of design can help improve the understanding of target

users, increase product ergonomics and aesthetics, and create or even redesign the products associated services.

2.2 Experiment Design and Research Instruments

Methodologically, a mixed-methods design was employed, in which the collection and analysis of quantitative and qualitative data were systematically been combined and triangulated. Pre-post surveys were employed as the quantitative method, whereas focus group interviews were conducted as the qualitative method. Gorard (2010) advocates that ‘mixed methods, in the sense of having a variety of tools in the toolbox and using them as appropriate, is the only sensible way to approach research’; the synthesis of quantitative and qualitative elements in a single project allows this study to minimize these deficiencies and limitations by allowing the data to be triangulated and complemented (Bryman, 2006; Greene et al., 1989).

With regard to the experimental design, this study adopted a Pretest and Posttest quasi-experimental design throughout an academic semester that lasted for 18 weeks. In order to have a more comprehensive understanding of the impact of the STEM IPBL course on the student’s learning and creativity, two research instruments with great measurement properties were used for data collection and analysis, including (a) Wu and Cherng’s (1992) ‘Motivated Strategies for Learning Questionnaire (MSLQ)’ was used to evaluate the participant’s perception of learning motivation (Cronbach’s $\alpha = .831$). MSLQ has three subscales, including “self-efficacy” (Cronbach’s $\alpha = .836$), “joyfulness of learning” (Cronbach’s $\alpha = .725$), and “valuing the significance of learning on future career development” (Cronbach’s $\alpha = .824$). (b) Chen’s (2006) ‘Abbreviated Torrance Test for Adults (ATTA)’ was used to measure the participant’s creativity. The ATTA has good measurement properties: test-retest reliability (.34~.68), internal-rater reliability (.31~.97), and criterion validity (.37~.46). The ATTA was used

to quantify the participant's creative strengths in four facets, including fluency, flexibility, originality, and elaboration through figural and verbal tests.

In the qualitative part of the study, focus group interviews were used to investigate the participants' perceptions towards the practice of the course, which also helped to triangulate the quantitative findings. More specifically, based on the participants' willingness, four group interviews that lasted for approximately 60 minute were conducted, each group was consisted of five participants randomly selected from the sample. Some questions were asked in the interviews, such as "What have you learned from the working and collaboration with students of different majors", "Based on your observation, what were the differences between the students from different majors?", and "what was the impact of the newly developed course on your learning motivation, creativity, and knowledge acquisition?" In addition, in order to be more objective in the evaluation of the participant's projects, their final products were consensually evaluated by a wide range of stakeholders, including the professionals of the fields, industrial practitioners, target users, and peer students.

2.3 Course Design

The STEM IPBL course aimed to cultivate the college student's talents of applying interdisciplinary skills and knowledge into their HCI projects, which were developed with an altruistic purpose of solving real-world problems encountered in the daily basis. The course was organized by three professors with diverse professional backgrounds. One professor's background covers the disciplines of industrial design, information science, and psychology, one professor's professional background includes computer science and information engineering, whereas one professor's professional background

consists of creativity and imagination education, educational psychology, guidance and counseling, and educational policy analysis.

As shown in Table 1, the elements of Science(S), Technology(T), Engineering(E), and Mathematics(M) were systematically and strategically embedded in the 18-week course. It should be noted that in order to enable the students to make the most use of the expertise and skills obtained from the course, the course coordinators additionally provided several lectures/workshops to empower the student's ability to integrate the domain knowledge of STEM into HCI systems development.

Insert TABLE 1. here

2.3 Course Deliver

The STEM IPBL course was delivered through the four phases of Design Thinking approach (IDEO, 2003; Brown, 2009). The first phase was “*Discover*” (from Week 2 to Week 5): the students were encouraged to use empathy to discover the real-world issues of their interests. The second phase was “*Define*” (from Week 6 to Week 8): the students were asked to clearly articulate the specific problem they would like to solve by applying STEM knowledge and skills acquired throughout the course. The third phase was “*Develop*” (from Week 9 to Week 15): the most important task the students needed to accomplish was to brainstorm the potential solutions for dealing with the specific problem identified in the previous phase. Before entering the final “*Deliver*” phase, a card deck named “Design with Intent (DWI) toolkit (Lockton et al, 2008; Lockton et al, 2010a; Lockton et al, 2010b)” was utilized to stimulate the participant's ideation. The DWI toolkit provides real-world examples of behavior design which are

drawn from many different disciplinary pieces of knowledge related to behavior change. It is useful in moderating the disciplinary differences in the ideation process (Kuo, Tseng, and Chang, 2018). Moreover, design the interactive system to foster particular target behavior is a higher goal of the students' projects. The project in this course is based on a guideline - to develop students' ability to solve real social problems and change society. Although many card-based design ideation tools (Wölfel & Merritt, 2013) were developed to help designers generate more ideas, only DWI is designed to generate ideas that promote behavioral change. Therefore, the DWI was used as the ideation tool in the project."

The final phase was "*Deliver*" (from Week 16 to Week 18), the HCI systems developed by each group were presented to a wide range of potential users, including professionals, teachers, industrial practitioners, and peer students. In this phase, we particularly invited two background experts to review the students' works. One background is related to HCI design experts, and the other is HCI technology-related experts. Two of the three design background experts came from academia and one from industry. All three of them are regular reviewers of design works and also have more than five years of experience in evaluating works. Experts in the technical background also come from academia and industry. Two people from academia and one from industry. They have more than five years of experience in developing HCI systems, and all have more than three years of review experience. Each student team must present their work to two review teams, separately. One presentation focused on the development of design, and the other focused on technological development. Other stakeholders were free to participate and they could give feedback to students at any time. 7 criteria were used for reviewing participants' design works. They are originality, novelty and creativity, technical depth, quality of test, completeness, fun, and quality of oral. These criteria

were often used in the review of HCI related projects. The review focused on giving students feedback on the project. In the project review, giving students direct feedback helped students understand the strengths and weaknesses of their projects and their future direction. More specifically, at the end of the course, all the accomplished HCI systems/products were invited to participate in a competition that organized by the hosting professors. Based on the evaluation of the feasibility, creativity, and usefulness of the accomplished HCI systems, the competition awarded the 1st prize team with 1,600 USD, the 2nd prize team with 1,000 USD, and the 3rd prize team with 300 USD. In addition, in order to encourage more participants to continuously develop HCI systems, some awards/prizes were also given to the participated teams, including the most popular award, most creative award, and funniest award.

3. RESULTS

3.1 Accomplished Human-Computer Interaction projects

The completion of the STEM IPBL course can be seen as a bold initiative and contribution to the current study. The participants were initially grouped into 20 teams. But, in determining the direction of the project, some teams found that the direction of the other teams was very similar to that of the team. This course allowed two groups to be merged into one group. Finally, there were 6 merged groups. The remaining eight groups did not merge. In the end, there were 14 groups. They accomplished 14 HCI projects. All the projects had an altruistic purpose and successfully applied the interdisciplinary knowledge and skills of STEM. The student's HCI works covers a big variety of topics, some examples are as follows: (a) "SLiMTAB" was developed based on Deep Neural Network (Science and Mathematics), electronic circuits (Engineering), and web development technology (Technology). By playing a song through the newly developed SLiMTAB system, the system can automatically generate sheet music

accurately for the music lovers or enthusiastic. (b) SVIM is a wearable device created for people with visual impairment to enjoy swimming. Combining the wireless local positioning technology and the pose estimation algorithm based on the Initial Measurement Unit (Science, Technology, Engineering, and Mathematics), the system can infer the head pose of the blind swimmers and give vocal instructions to avoid collision and deviation while enjoying swimming. (c) “PosBet” was developed based on Initial Measurement Unit and movement evaluation algorithm (Science, Technology, Engineering, and Mathematics). The system was created for the patients suffering from an occupational injury to have better knowledge of health care and rehabilitation.

Other HCI systems were also developed through the use of STEM knowledge and skills. Some were created for behavior design, for example, “Greender” is a mobile app designed to help the public to do waste classification/sorting; “Parker” is an app developed for the target users to do more exercises in their leisure time; whereas “Deli City” is an app developed to help the target users to explore the gourmet restaurant based on the user’s preferences and location. Some systems were designed for a very specific group of target users, such as “ICD 10 search assistant” was developed to help medical-relevant users to search for the correct ICD 10 and offer valuable information of the searched disease.

3.2 Learning Motivation

As shown in Table 2, the participant’s learning motivation was improved significantly ($p=.012$). Rating on the 6-point Likert MSLQ, the participants’ ratings increased from Pretest ($M = 4.40$, $SD = .39$) to Posttest ($M=4.64$, $SD = .49$). Engineering majored students (Pre: $M = 4.35$, $SD = .41$; Post: $M = 4.65$, $SD = .53$; $p=.02$) experienced better improvement than design majored students (Pre: $M = 4.49$, $SD = .35$ Post: $M = 4.63$;

SD = .42; $p=.33$). More specifically, in the MSLQ the student's "self-efficacy", "joyfulness of learning", and "recognizing the significance of learning on future career development" were also measured, the results are highlighted in the following sections from 3.2.1 to 3.2.3.

Insert TABLE 2. here

3.2.1 Self-Efficacy

The students were asked some questions considering their self-efficacy of learning interdisciplinary knowledge and skills, such as "I am sure that I can master all the knowledge and skills taught in the course", "I am sure that I understand the most difficult knowledge and skills taught in the course", and "I am confident that I can have great performance in the final test and projects". As shown in Table 2, in the 6-point Likert questionnaire, the participant's self-efficacy improved significantly ($p= .003$) from Pretest (M = 4.03; SD = .64) to Posttest (M=4.43; SD = .61). Both engineering majored students (Pre: M = 4.05, SD = .65; Post: M = 4.43, SD = .68; $p= .029$) and design majored students (Pre: M = 3.99, SD = .64; Post: M = 4.41, SD = .44; $p=.044$) witnessed a significant improvement in their self-efficacy of learning. For instance, a leader of one the groups reported that:

"We are very surprised [about our final projects]! We never thought that we can make it (the development of HCI system) happen. Working with people with divergent backgrounds has its value, which makes it easier to master [knowledge and skills of] different domains, now, they (students with design majors) can say they understand engineering, coding, and deep learning, and we (engineering majored students) can say that we know the mystery of design."

3.2.2 Joyfulness of Learning

Two questions were asked to investigate the student's joyfulness of learning, including "I am very interested in learning the contents taught in the course" and "I like the course design and how it is delivered." Evidence indicated that the participant's perceptions on the joyfulness of learning through STEM IPBL course improved from Pretest ($M = 4.68$, $SD = .44$) to Posttest ($M=4.75$, $SD = .61$), yet, no statistical significance yielded ($p=.556$). However, it should be noted that while engineering majored student's joyfulness of learning improved (Pre: $M = 4.58$, $SD = .44$; Post: $M = 4.77$, $SD = .62$; $p=.178$), the ratings of design majored students were marginally declined (Pre: $M = 4.89$, $SD = .39$; Post: $M = 4.71$, $SD = .62$; $p=.353$). Engineering majored students argued that it is interesting to learn the expertise of design. For instance, a group of students reported:

"it is enjoyable to discuss and work with people from different backgrounds, knowing design thinking has a great impact on our project, and we have lots of meaningful discussions with the design [majored students], as they always come out with imaginative ideas."

However, some design majored students expressed that they experienced numerous difficulties in the acquisition of engineering knowledge and skills, a group of design majored students reported:

"Although the whole [course] design is great, working with them(engineering majored students) is great and we like the collaboration, we still experienced an awkward feeling that for engineering students, comparatively, it seems easier for them to understand our domain [knowledge and skills], however, some of us [with design majors] don't even understand their symbols (the symbols used by engineering majored students), how can we [do the sort of task of] coding, which [makes it] difficult for us to immerse, or you can say enjoy ourselves in the course, yet, it is still a great course that we would recommend others to take."

3.2.3 Recognizing the Significance of STEM IPBL Learning on Future Career

As shown in Table 2, a significant growing value of the importance of the STEM IPBL course on future career development was found from comparing the results of Pretest ($M = 4.73$, $SD = .62$) and Posttest ($M = 4.94$, $SD = .52$) ($p = .077$). Among the participants, engineering majored students (Pre: $M = 4.62$, $SD = .69$; Post: $M = 4.92$, $SD = .54$; $p = .066$) experienced better improvement than design majored students (Pre: $M = 4.93$, $SD = .36$; Post: $M = 4.98$; $SD = .5$, $p = .781$). The results can be interpreted that most students from both engineering and design backgrounds were well-recognized the significance of the interdisciplinary knowledge, skills, and collaboration on their future career. However, the values of engineering majored students grew more than design majored students. For example, a group of students reported

“For all of us [participating in the course] the course has reminded or you can say enabled us to pay more attention to what the potential users want, including the customers and the industry. Inviting people from outside (the representativeness of industries who were invited to review the student’s final projects) is a wise move, which makes it (the course) very different [from other course], as the people [from industries] told us what kind of people they want to recruit, what kind of expertise is required [to be successful] in the future job seeking”

3.3 Creativity

Insert TABLE 3. here

The student’s overall creativity was computed as a summation of four facets of creativity, including fluency, originality, flexibility, and elaboration. The student’s raw scores on the four facets of creativity were transformed into standard scores, while the highest ranking is 20, the lowest ranking is 0. That is, the highest standard score of overall creativity is 80, the results indicated that student’s overall creativity was

improved from Pretest to Posttest (Pre M= 63.36, SD = 4.57; Post M= 68.44, SD = 3.35; $p=.000$).

The participant's fluency improved significantly (Pre M= 14.89, SD = 1.79; Post M= 16.20, SD = 1.8; $p= .001$), which indicates that they can come out with more ideas within a limited period of time. The participant's elaboration increased significantly (Pre M= 16.69, SD = 1.47; Post M= 18.62, SD = .78; $p= .000$), which refers that the participants can increase the sophistication of their projects, such as adding more details, creating new meanings or interpretations of things, or refining their ideas for putting into practice. The participant's flexibility has improved as well (Pre M= 14.82, SD = 2.33; Post M= 16.04, SD = 1.97; $p= .009$), this indicates that the participants can make more connections between ideas, come out with alternatives with a great diversity. While engineering majored student's flexibility increased from Pre M=14.33, SD =2.01 to Post M= 15.7, SD=1.97 ($p=.01$), design majored student's flexibility increased from Pre M=15.80; SD =2.68 to Post M= 16.73, SD=1.83 ($p=.275$). It is reasonable to see that even though engineering majored students enjoyed a better improvement, design majored students performed better in both Pretest and Posttest, as most of them have better trained to think in a more flexible basis and think more divergently.

It is found that the STEM IPBL course had different impact on the student's originality, while the engineering majored student's originality enjoyed a significant improvement (Pre M= 16.77, SD = 2.01; Post M= 17.93, SD = .98; $p=.006$), the design majored student's originality witnessed a slight decrease, though no statistical significance was yielded (Pre M= 17.33, SD = 1.54; Post M= 16.87, SD = 3.23; $p=.617$). The results drawn from interviews with the students revealed that the interdisciplinary collaboration can really help the participants to come out with more creative ideas, and

further develop or refine their design concepts. However, in order to put their enormous ideas into practice, some ideas that were being ‘novel’ but not ‘appropriate’, or imaginative needed to be eliminated, because eventually the ideas must be implemented/produced as prototypes for later evaluation. For example, a group of engineering majored students expressed:

“They (design majored students) are extremely creative, sometimes their ideas are too imaginative so that we need to drag them [from their imaginative thoughts] to reality, letting them know that we have our technical limitations, we cannot turn everything into reality, we don't have unlimited resources!”

Another interesting quote was from a group of design majored students:

“We (design majored students) know we have been too wild, but you know, this is what we do, and we need to keep doing this in our lifetime if we want to pursue a career that is called design. It's fun to work with them (engineering majored students), it's good to see some of our ideas that can be produced [in reality], you know, we act the role of Steve Jobs, not all of our ideas can be put into practice!” and “[We are (design majored)] too original for them (engineering majored)! We (design majored students) need to slow down a little bit, otherwise, they (engineering majored students) cannot follow...We need to be more realistic, because impossible is impossible, [we need to] adjust ourselves and [be more] thoughtful (to abandon some ideas that are too original or imaginative) for them (engineering majored students) in the product design process...”

That is, the efforts of refinement and tailoring were argued by the participants as a hinder factor leading to the marginal decline of their originality, though, no statistically significant difference was yielded from the results of Pretest and Posttest.

4. DISCUSSION

4.1 STEM IPBL can be regarded as a feasible approach to promote STEM and PBL education for HCI systems development

The practice of STEM IPBL course was based on past scholarly works, as the literature pointed to three important directions that guide this study: Firstly, STEM is the skill set that is “expected to be held by people with a tertiary-education level degree in the subjects of science, technology, engineering and mathematics” (EU Skills Panorama Glossary, 2014; Fans and Ritz, 2014). STEM education and creativity are interconnected and inter-influenced, as creativity is the fundamental sauce in the success in STEM learning (Turnipseed, 2013; Ramirez, 2013), and STEM education can help promote the learner’s creativity (Craft et al, 2014; Eguchi, 2016; Land, 2013; Madden et al, 2013). Therefore, to strategically embed S, T, E, and M into college engineering and design education has its significance and appropriateness, as it not only offers the students essential knowledge and skill sets but also has a potential to improve their creativity.

Secondly, the development of HCI system often requires interdisciplinary knowledge, such as design and technology (Te’eni et al, 2007; Preece et al, 2015; Hartson, 1998; Hewett et al, 1992) and interdisciplinary team collaborations (Duncan et al, 2014; Lathia et al, 2013; Stawarz, 2015). The interdisciplinary collaboration can help optimize the requirements of desirability, feasibility, and viability (Kandachar, 2010; Smith, 2007) for developing products, services, and technologies for the specific purposes. Therefore, the approach used by the current study to merge students from different backgrounds (ID, CSIE, and EE) to form teams on HCI systems development can be seen as an attempt to enable the students to make contributions for a common task by applying the knowledge and skills from their own domains. Furthermore, recent

works indicate that the inclusion of diverse groups in STEM education can lead to the increase of the student's creativity and innovation (McBride, 2017).

Thirdly, PBL is a very effective approach to immerse the learners into a contextualized and authentic learning setting, has therefore been commonly utilized to nurture the learner's creative problem-solving capability in dealing with real-world problems (Grant and Branch, 2005; Hsu, 2004; Scott et al, 2004). Furthermore, it can benefit the learner's acquisition and application of 21st-century skills, including higher order thinking, cooperation and collaboration, and communication and presentation skills. Therefore, in order to empower the students to come out with creative ways to solve real-world issues by making HCI systems, PBL approach is utilized by the current study.

It also need to be noted that in the past years of teaching and conducting research in higher education institutions, it is found that even though the engineering majored students have great capability to develop well-functioned systems, some of them have encountered with numerous obstacles, some of them do not know how to conduct user study, have not been trained to improve the aesthetics of their products. Thus, in the recent years, rather than doing STEM education, there is an increasing number of scholars and professionals advocate so-called STEAM education, which include the element of Art into STEM education, as art can be regarded as a nutritious adjustment to the STEM education (Burnard et al, 2017; Land, 2013; Madden et al, 2013). The current study is conducted in a top-tier university in the country, it recruits a group of the most intellectual students to study in different disciplines. It also makes it possible to merge students from engineering and design departments to participate in the same course. It is believed that design majored students have abundant experience of conducting user study, including doing a pre-user study to know the needs of potential

users, and post-user study to evaluate the adequacy, usefulness, ergonomic, and aesthetics of the products. The inclusion of design majored students in a HCI project not only makes STEM become STEAM but also contribute to a better understanding of the users and evaluation of the final products.

4.2 Impact of STEM IPBL on student's learning motivation

The data collected through the 18-week course indicated that the 45 college students have experienced great improvements on the learning of interdisciplinary knowledge and skills so that all the 20 teams made significant contributions on the accomplishments of 14 HCI projects (as highlighted in 3.1 above). The results support the literature that the STEM IPBL course can empower the students from divergent backgrounds to participate in interdisciplinary collaboration and help develop better products. Evidence indicated that the course improved the student's learning motivation and creativity significantly.

More specifically, the evidence drawn from a comparison of Pretest and Posttest results indicate that the intervention had a great impact on the student's overall learning motivation (Pre $M = 4.40$, $SD = .40$; Post $M = 4.64$, $SD = .49$; $p = .012$), which had better effect on engineering majored ($p = .02$) than design majored ($p = .33$). The variable of overall learning motivation is the summation of its three subscales, including self-efficacy, the joyfulness of learning, and recognizing the significance of learning on future work. The best effect was found in the improvement of the student's learning efficacy (Pre $M = 4.03$, $SD = .64$; Post $M = 4.43$, $SD = .61$; $p = .003$). Both engineering majored students ($p = .029$) and design majored students ($p = .044$) enjoyed a great improvement in their self-efficacy of learning. The results indicated that the students

have become more confident, believing they can learn the knowledge and skills taught in the course more efficiently and then apply to system development.

With regard to the joyfulness of learning, although an increase was found by comparing the student's Pretest and Posttest response (Pre $M = 4.68$, $SD = .44$; Post $M = 4.74$, $SD = .61$), no statistical significance was yielded ($p = .556$). A more detailed examination of the student's responses indicated that while the engineering majored students more enjoyed in the course (Pre: $M = 4.58$, $SD = .44$; Post: $M = 4.77$, $SD = .62$; $p = .178$), some design majored students held different ideas (Pre: $M = 4.89$, $SD = .39$; Post: $M = 4.71$, $SD = .62$; $p = .353$). Some possible reasons were found in the interviews: firstly, to engineering majored students, it's comparatively easier for them to learn the knowledge of design, but for design majored students, although the discussion and collaboration were regarded as meaningful and insightful, some of them have had difficult time learning the expertise of engineering discipline. Secondly, the results were computed by comparing Pretest and Posttest scores, although there was a marginal decline of the design majored student's responses, their ratings were, still, quite positive in the Posttest (4.71 out of 6). The results might due to the student's expectation was too high in the beginning, or in the course, the engineering majored students have played the more important role than the design majored students in the later part of HCI system development, some of the design majored ones thus felt less-participated in the Posttest period of time. It was also found that most students from both engineering and design backgrounds were well-recognized the importance of the interdisciplinary knowledge, skills, and collaboration on their future career Pre $M = 4.73$, $SD = .62$; Post $M = 4.94$, $SD = .52$; $p = .077$).

4.3 Impact of STEM IPBL on student's creativity

With regard to the changes in the student's creativity throughout the intervention of STEM IPBL course, the analysis of data collected from Pretest and Posttest of TTAT indicated a pleasant improvement. The student's overall creativity improved significantly from $M=63.36$, $SD=4.57$ to $M=68.4$, $SD=3.35$ ($p=.000$). A detailed examination of the four facets of creativity indicated that both engineering and design majored student's fluency (Overall $p=.001$; design $p=.013$, engineering $p=.021$) and elaboration (overall $p=.000$; design $p=.000$, engineering $p=.000$) improved significantly. The results showed that after participating in the course the students can think faster, come out with more ideas, and have the ability to put more details into their ideas.

The course had a better effect on engineering majored student's flexibility and originality than design majored ones. In overall, the participant's flexibility improved (Pre $M=14.82$, $SD=2.33$; Post $M=16.04$, $SD=1.97$; $p=.009$), this indicated that the participants can make more connections between ideas, come out with alternatives with a great diversity. While engineering majored student's flexibility increased from Pre $M=14.33$, $SD=2.01$ to Post $M=15.7$, $SD=1.97$ ($p=.01$), design majored student's flexibility increased from Pre $M=15.80$; $SD=2.68$ to Post $M=16.73$, $SD=1.83$ ($p=.275$). It is reasonable to see that even though engineering majored students enjoyed a better improvement, design majored students performed better in both Pretest and Posttest, as most of them have been better trained to think in a more flexible basis and think more divergently. However, the results indicated that the design and delivery of the course have had its value in broadening the student's horizons, empowering them to think in a more diverse manner.

It is found that the STEM IPBL course had different impact on the student's originality, while the engineering majored student's originality enjoyed a significant improvement

(Pre $M = 16.77$, $SD = 2.01$; Post $M = 17.93$, $SD = .98$; $p = .006$), the design majored student's originality witnessed a slight decrease, though no statistical significance was yielded (Pre $M = 17.33$, $SD = .99$; Post $M = 16.87$, $SD = 3.23$; $p = .617$). The results alarm us that although most of the students expressed their appreciation of interdisciplinary collaboration and discussion, the teachers/lecturers need to remind the participants to treasure, or at least not to sacrifice their so-called "most original" ideas. It is true that some design majored students reported that in the system development process, some of them needed to become more "realistic", to give up some ideas that were too original, creative, or difficult to be implemented. That is, to refine or tailor their original ideas into an actual product could be regarded as a hinder factor leading to the marginal decline of their originality. However, for engineering majored ones, the results indicated that working with fellow students with different majors can positively impact their originality, enable them to come out with more unique and imaginative ideas.

4.4 Limitations

Given there is a possibility to add an equivalent group of participants, it would be beneficial for the potential comparison of experimental group and control group. The comparison of the pre-post performance of the two groups of participants can be helpful in the identification of impact made by the intervention, however, two reasons led us to applied one group pre-post design: (1) the STEM IPBL course is a brand new course taught in the university (where the study is conducted), and there are no other similar courses taught at the same period of time. (2) in natural educational settings, it is difficult to find so-called "equivalent group" that has similar baseline behavior and performance. Thus, due to the constraints like the limited budgets and resources, a one group pre-post design was used by the current study. We well-noted the limitation that is inherent from the experimental design employed by the current study, we thus

recommend future studies can consider to apply a pretest-posttest equivalent-group design given there is sufficient budget, resources, and a possibility to recruit a control group with similar baseline behavior, performance, and motivation to participant in the study.

5. CONCLUSION

The initiative of STEM IPBL course can be regarded as one of the frontiers in engineering education and an evidence-based practice. The course systematically embedded the domain knowledge of STEM and successfully merged the participants from different majors. Following the procedures suggested by Design Thinking, the course delivered through four phases (Discover, Define, Develop, and Deliver, which organized by three professors with distinctive professional backgrounds, and also taught by a wide range of professionals, and industrial practitioners/partners. The STEM IPBL has been indicated as a very effective approach to empower engineering and design students to develop HCI systems. Throughout the experimental period, the course improved the participants' (a) learning motivation ($p = .012$), (b) self-efficacy ($p = .003$), (c) joyfulness of learning ($p = .556$), and (d) recognizing the significance of learning on future work ($p = .077$). The students also witnessed a great improvement on creativity ($p = .000$), among the four facets of creativity, the participants' fluency ($p = .001$), elaboration ($p = .00$), and flexibility ($p = .009$) were improved significantly. However, while engineering majored student's originality increased significantly ($p = .006$), design majored student's originality was marginally declined ($p = .617$). Some design majored students reported that some of their ideas might be "too original" for those engineering majored students, which may be very difficult to be implemented in the production of HCI systems, especially when worked in a team of members with divergent backgrounds.

It is an earnestly hope that the initiative of STEM IPBL course, delivered through an intriguing and inspiring teaching strategy, can make a constant contribution in engineering education and be extensively applied to ignite the passion and interests of college students and future learners in engineering study and better their future career development.

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REFERENCES

- Agapie, E., Avrahami, D., and Marlow, J. (2016). Staying the Course: System-Driven Lapse Management for Supporting Behavior Change. In Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems, ACM, 1072-1083.
- Burnard, P., Dragovic, T., Jasilek, S., Fenyvesi, K., Rolls, L., Durning, A., and Biddulph, J. (2017) 'The art of creating possibility spaces for fostering STEAM practices in primary education'. In X. Du and T. Chemi (Eds) *Innovation and Change in Education Cross-cultural Perspectives Special Issue: Arts-based Methods in Education Across the World*. River Publishers.
- Brown, T. (2009). *Change By Design*. HaperCollins Publisher. [LSEP]
- Bryman, A. (2006). Integrating quantitative and qualitative research: How is it done? *Qualitative Research*, 6(1), 97-113.
- Chen, C. Y. (2006). *Abbreviated Torrance Test for Adults manual* in Chinese Version. Taipei: Psychology Publisher. (In Chinese).
- Consolvo, S., McDonald, D. W., and Landay, J. A. (2009). Theory-driven design strategies for technologies that support behavior change in everyday life. in Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, ACM, 405-414.
- Craft, A., Hall, E., & Costello, R. (2014). Passion: Engine of creative teaching in an English university? *Thinking Skills and Creativity*, 13, 91–105. <https://doi.org/10.1016/j.tsc.2014.03.003>
- Cropley, D. H. (2016). Creativity in engineering. In *Multidisciplinary Contributions to the Science of Creative Thinking*, Springer Singapore, pp. 155-173.
- Duncan, L. R., Hieftje, K. D., Culyba, S. and Fiellin, L. E. (2014). Game playbooks: tools to guide multidisciplinary teams in developing videogame-based behavior change interventions. *Translational behavioral medicine*, 4(1), 108-116.
- Dym, C. L., Agogino, A. M., Eris, O., Frey, D. D., & Leifer, L. J., (2013). Engineering design thinking, teaching, and learning. *Journal of Engineering Education*, 94(1), 103-119.
- Eguchi, A. (2016). RoboCupJunior for promoting STEM education, 21st-century skills, and technological advancement through robotics competition. *Robotics and Autonomous Systems*, 75, 692–699. <https://doi.org/10.1016/j.robot.2015.05.013>
- EU Skills Panorama (2014) STEM skills Analytical Highlight, prepared by ICF and Cedefop for the European Commission
- Fans, S. & Ritz, J. M. (2014). International views of STEM Education.
- Fogg, B. J. (2002). Persuasive technology: using computers to change what we think and do. *Ubiquity*, 2002 (December).

- Gorard, S. (2010). Research design, as independent of methods. In A. Tashakkori & C. Teddlie (Eds.), Sage handbook of mixed methods in social & behavioral research. London, UK: Sage Publications.
- Grammenos, D., & Antona, M. (2017). Future designers: Introducing creativity, design thinking & design to children. *International Journal of Child-Computer Interaction*. <https://doi.org/10.1016/j.ijcci.2017.10.002>
- Grant, M. M. & Branch, R. M. (2005). Project-Based Learning in a middle school : Tracing abilities through the artifacts of learning. *Journal of Research on Technology in Education*, 38(1), 65-98.
- Greene, J. C., Caracelli, V. J., & Graham, W. F. (1989). Toward a conceptual framework for mixedmethod evaluation designs. *Educational Evaluation and Policy Analysis*, 11(3), 255-274.
- Hartson, H. Rex. (1998) Human-computer interaction: Interdisciplinary roots and trends, *Journal of Systems and Software*, Volume 43, Issue 2, pp. 103-118, ISSN 0164-1212.
- He, H. A., Greenberg, S., and Huang, E. M. (2010). One size does not fit all: applying the transtheoretical model to energy feedback technology design. in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, ACM, 927-936.
- Hekler, E. B., Klasnja, P., Froehlich, J. E., and Buman, M. P. (2013). Mind the theoretical gap: interpreting, using, and developing behavioral theory in HCI research. in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, ACM, 3307-3316.
- Hewett, T. T., Baecker, R., Card, S., Carey, T., Gasen, J., Mantei, M., Perlman, G., Strong, G., and Verplank, W. (1992). *ACM SIGCHI Curricula for Human-Computer Interaction*. Technical Report. ACM, New York, NY, USA.
- Hirsch, P. L., Shwom, B. L., Yarnoff, C., Anderson, J. C., Kelso, D. M., Olson, G. B., & Colgate, J. E. (2001). Engineering design and communication: The case for interdisciplinary collaboration. *International Journal of Engineering Education*, 17(4/5), 343-348.
- Hsu, S. (2004). Using case discussion on the web to develop student teacher problem-solving skills. *Teaching and Teacher Education*, 20(7), 681-692.
- IDEO. (2003). *Method cards: 51 ways to inspire design*. Palo Alto.
- Calvo, I., Cabanes, I., Quesada, J., & Barambones, O. (2018). A Multidisciplinary PBL Approach for Teaching Industrial Informatics and Robotics in Engineering. *IEEE Transactions on Education*, 61(1), 21-28.
- Kandachar, P. (2010). Designing for global sustainable solutions. in F. Ceschin, C.V.a.J.Z. ed. *Sustainability in Design: Now! Challenges and Opportunities for*

- Design Research, Education, and Practice in the XXI Century, Greenleaf Publishing Limited, London, 60-75. ^[1]_{SEP}
- Klasnja, P., Consolvo, S., and Pratt, W. (2011). How to evaluate technologies for health behavior change in HCI research. in Proceedings of the SIGCHI
- Kuo, H. C., Tseng, Y. C., and Chang, L. C. (2018). *Promoting Design, Humanities, and Technology Majored Students' Design Ideation through Design with Intent (DWI) Cards*. British Education Research Association (BERA) International conference.
- Lafrenière, D., Dayton, T., and Muller, M. (1999). Variations of a theme: card-based techniques for participatory analysis and design. in CHI'99 Extended Abstracts on Human Factors in Computing Systems, ACM, 151-152. ^[1]_{SEP}
- Land, M. H. (2013). Full STEAM Ahead: The Benefits of Integrating the Arts Into STEM. *Complex Adaptive Systems*, 20, 547-552. <https://doi.org/10.1016/j.procs.2013.09.317>
- Lockton, D., Harrison, D., and Stanton, N. (2008). Design with intent: Persuasive technology in a wider context. in Persuasive technology, Springer, 274- 278. ^[1]_{SEP}
- Lockton, D., Harrison, D., and Stanton, N. (2010a). The Design with Intent Method: A design tool for influencing user behavior. *Applied Ergonomics*, 41 (3). 382-392.
- Lockton, D., Harrison, D., and Stanton, N. (2010b). Design with intent: 101 patterns for influencing behavior through design. *Equifine*. ^[1]_{SEP}
- Lockton, D., Harrison, D. & Stanton, N. A. (2010). The Design with Intent Method: A design tool for influencing user behavior. *Applied Ergonomics*, 41(3), 382-392.
- Madden, M. E., Baxter, M., Beauchamp, H., Bouchard, K., Habermas, D., Huff, M., ... Plague, G. (2013). Rethinking STEM Education: An Interdisciplinary STEAM Curriculum. *Complex Adaptive Systems*, 20, 541–546. <https://doi.org/10.1016/j.procs.2013.09.316>
- Mamykina, L., Candy, L., & Edmonds, E. (2002). Collaborative creativity. *Communications of the ACM*, 45(10), 96-99.
- McBride, L. (2017). Inclusion of Diverse Groups in STEM Leads to Increased Creativity, Innovation. Insight into Diversity. Retrieved on April 4, 2018, <http://www.insightintodiversity.com/inclusion-of-diverse-groups-in-stem-leads-to-increased-creativity-innovation/>
- Preece, J., Sharp, H., Rogers, Y. (2015). *Interaction Design: Beyond Human-Computer Interaction* (4th ed), Chichester: John Wiley & Sons Inc., ISBN: 1119020751. <https://arl.human.cornell.edu/879Readings/Interaction%20Design%20-%20Beyond%20Human-Computer%20Interaction.pdf>

- Ramirez, A. (2013). Creativity is the Secret Sauce in STEM. George Lucas Educational Foundation. Retrieved on April 4, 2018, <https://www.edutopia.org/blog/creativity-secret-sauce-in-stem-ainissa-ramirez>
- Saorín, J. L., Melian-Díaz, D., Bonnet, A., Carbonell Carrera, C., Meier, C., & De La Torre-Cantero, J. (2017). Makerspace teaching-learning environment to enhance creative competence in engineering students. *Thinking Skills and Creativity*, 23, 188-198. <https://doi.org/10.1016/j.tsc.2017.01.004>
- Scott, G., Leritz, L. E., & Mumford, M. D. (2004). The effectiveness of creativity training: A quantitative review. *Creativity Research Journal*, 16(4), pp. 361-388.
- Smith, S. (2007). Design for the Other 90%. Cooper-Hewitt, National Design Museum, Smithsonian Organization.
- Te'eni, D., Carey, J., and Zhang, P. (2007). Human-Computer Interaction: Developing Effective Organizational Information Systems, John Wiley and Sons Inc., ISBN: 0471677655.
- Turnipseed, S. (2013). The Importance of Creativity in STEM. 21 Partnership for 21st Century Learning. Retrieved on April 4, 2018, <http://www.p21.org/news-events/p21blog/1067-stephan-turnipseed>
- UKCES (2011) The supply and demand for high-level STEM skills, Briefing Paper, December 2011.
- UK Parliament (2012). Science and Technology Committee-Second Report: Higher education in Science, Technology, Engineering and Mathematics (STEM) subjects.
- Wu, J. J., and Cherng, B. L. (1992). Motivated Strategies for Learning Questionnaire(MSLQ): A revised version for use with Chinese elementary and junior high school students. *Psychological Testing*, 39, 59-78. (In Chinese).